

Real-Time DC-Link Capacitor Stress Reduction in Two-Level Three-Phase Inverters Using Adaptive Multi-Carrier PWM Method

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1 INTRODUCTION

- Two-level three-phase voltage source inverters are widely used in industrial motor drives.
- DC-link capacitors remain a major reliability bottleneck. Ripple current causes ESR heating, thermal stress, degradation, and reduced capacitor lifetime.
- Existing ripple-reduction methods often require multiple inverter units or additional converter stages.
- This work proposes adaptive interleaving among the three legs of a single inverter using real-time multi-carrier PWM, without adding extra hardware.

2 DC-LINK CAPACITOR LIFETIME ESTIMATION

- The capacitor lifetime is strongly affected by voltage stress and hotspot temperature.

$$L = L_{nom} \times \left(\frac{V_{op}}{V_{nom}} \right)^n \times 2^{\frac{T_{hot} - T_{nom}}{10}}$$

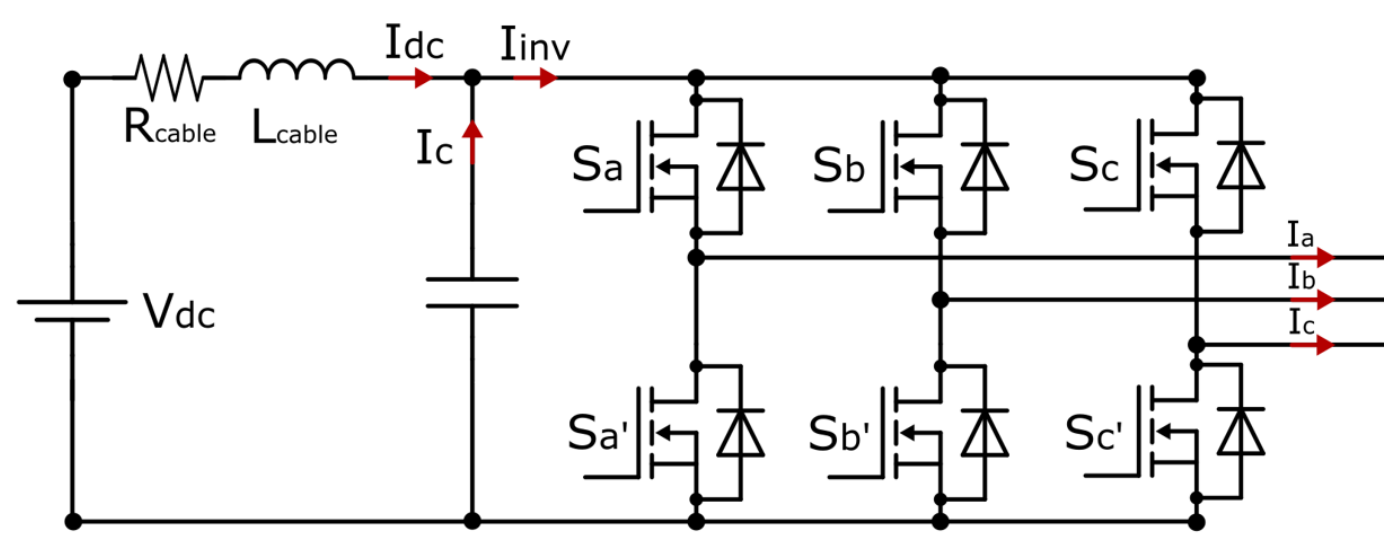
- Ripple-current-induced ESR losses increase the hotspot temperature, which accelerates degradation.

$$T^{hot} = T_{ambient} + R_{eq} \times P_{Loss-ESR}$$

- Therefore, reducing the RMS capacitor current directly reduces thermal stress and can extend the capacitor's lifetime.

$$P_{Loss-ESR} = \sum_{n=1}^{\infty} \left(I_{cap,RMS}^2(f_n) \times ESR(f_n) \right)$$

3 DC-LINK CURRENT ANALYSIS OF THE INVERTER



- Capacitor Current: $i_{cap}(t) = \sum_{x \in \{a,b,c\}} i_x(t) S_x(t) - \bar{i}_{dc}$

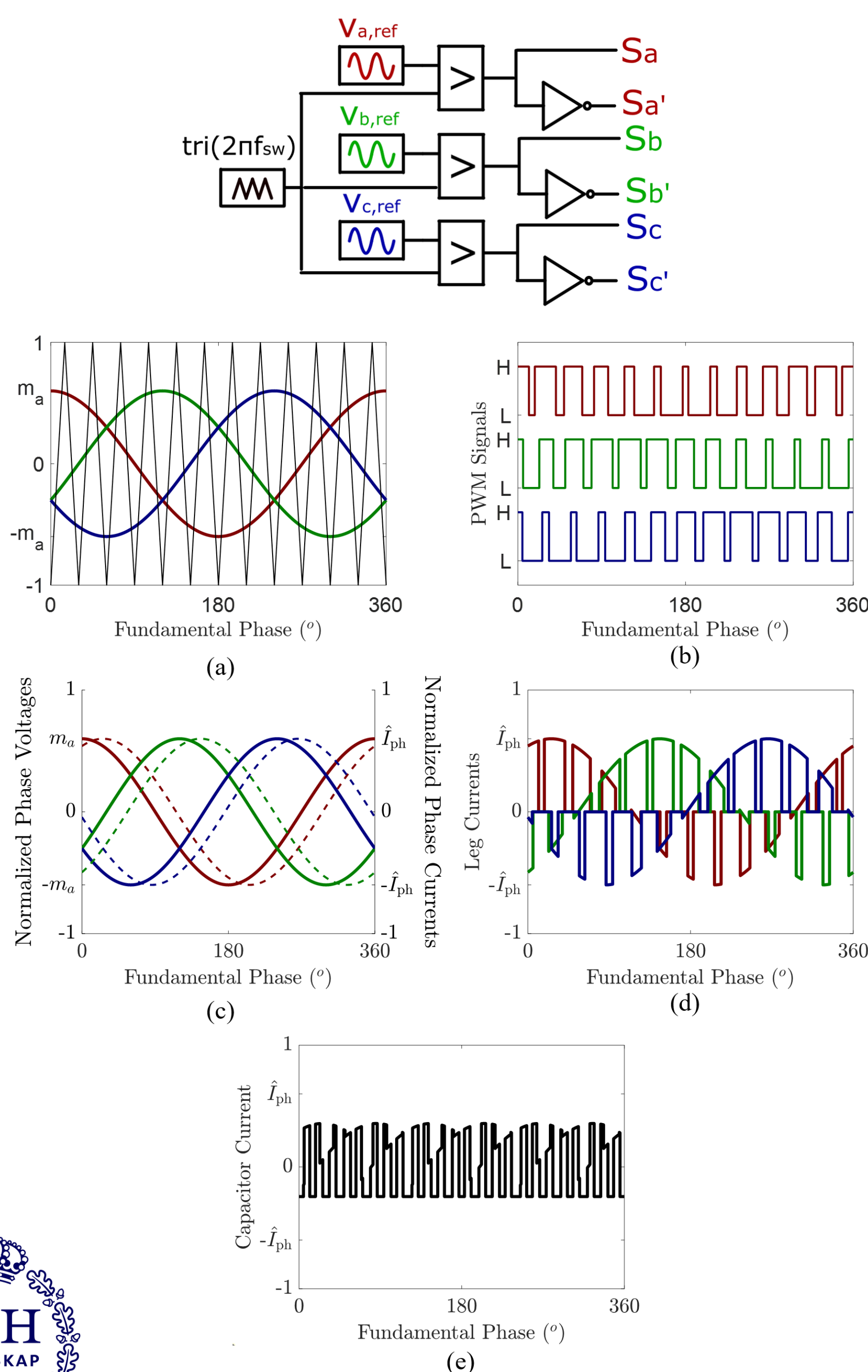
- RMS calculation within switching periods:

$$I_{cap,rms}[k] = \sqrt{\frac{1}{T_{sw}} \int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt} \quad \text{for } k = 1, 2, \dots, N_{sw}$$

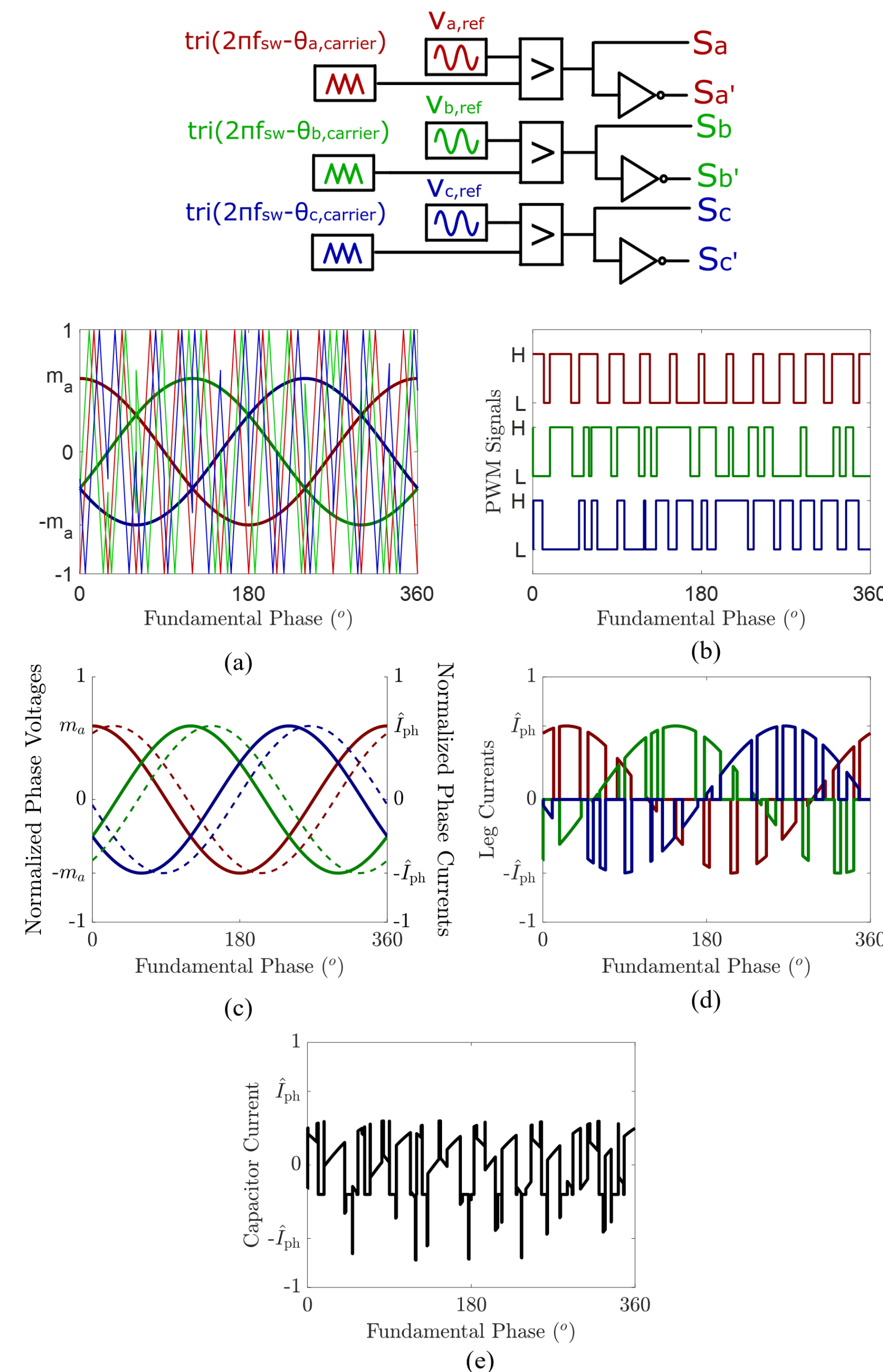
- Total RMS calculation :

$$I_{cap,rms}^{total} = \sqrt{\frac{1}{N_{sw}} \sum_{k=1}^{N_{sw}} I_{cap,rms}^2[k]}$$

4 CONVENTIONAL SINGLE-CARRIER-BASED PWM



5 PROPOSED MULTI-CARRIER PWM TECHNIQUE



6 IMPLEMENTATION METHODOLOGY

- Offline optimization & LUT-based real-time implementation
- Current normalization:

$$\hat{i}_x(t) = \frac{i_x(t)}{|i_a(t)| + |i_b(t)| + |i_c(t)|}, \quad x \in \{a, b, c\}$$

- Under the balanced three-phase condition, current inputs reduce to

$$i_a + i_b + i_c = 0 \quad \text{Current Inputs} \mapsto \{\hat{i}_a, \hat{i}_b\}$$

- Using the duty-cycle constraint in SPWM, duty inputs reduce to

$$d_a + d_b + d_c = \frac{3}{2} \quad \text{Duty Ratio Inputs} \mapsto \{d_a, d_b\}$$

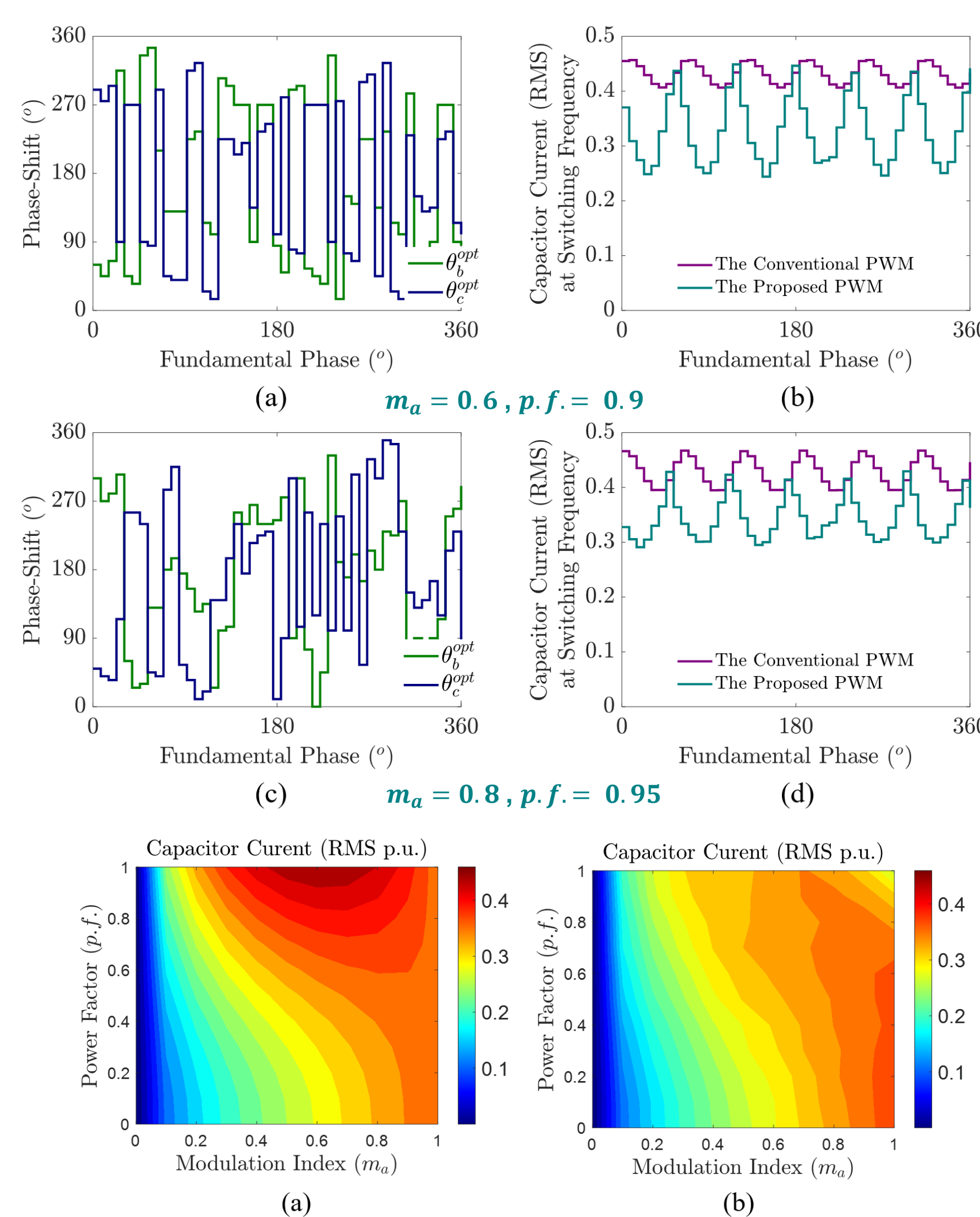
- Look-up-table mapping

$$\text{LUT} : (\hat{i}_a, \hat{i}_b, d_a, d_b) \mapsto (\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt})$$

- Optimization objective (per switching period k)

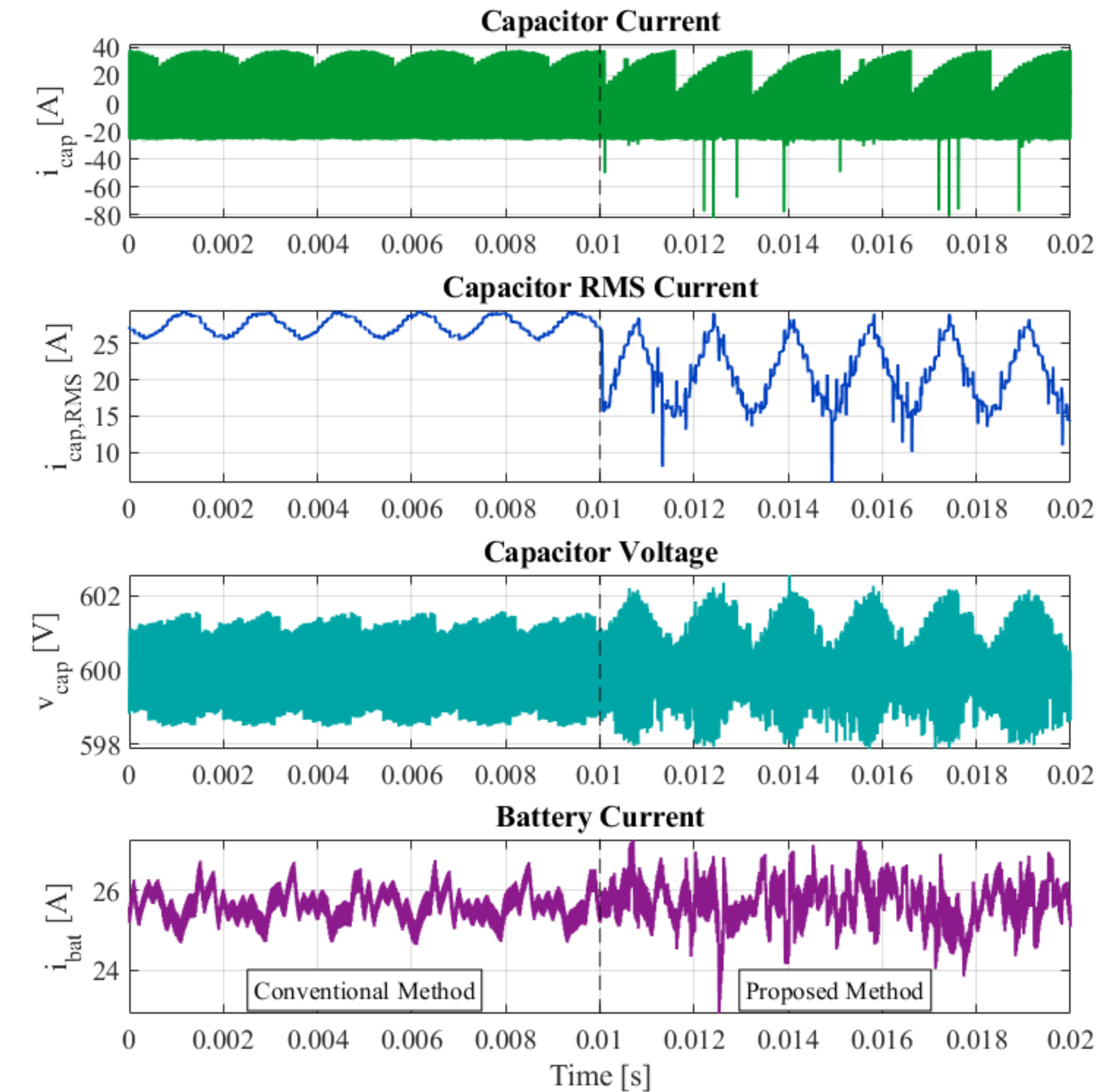
$$(\theta_{b,carrier}^{opt}[k], \theta_{c,carrier}^{opt}[k]) = \arg \min_{\theta_b, \theta_c} \left(\int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt \right)$$

7 PERFORMANCE ASSESSMENT



8 SIMULATION VALIDATION

Steady-state comparison ($m_a = 0.6$, $p.f. = 0.9$)



$$I_{cap,RMS} = 27.6 \text{ A} \quad I_{cap,RMS} = 21.2 \text{ A}$$

Reduction ~23%

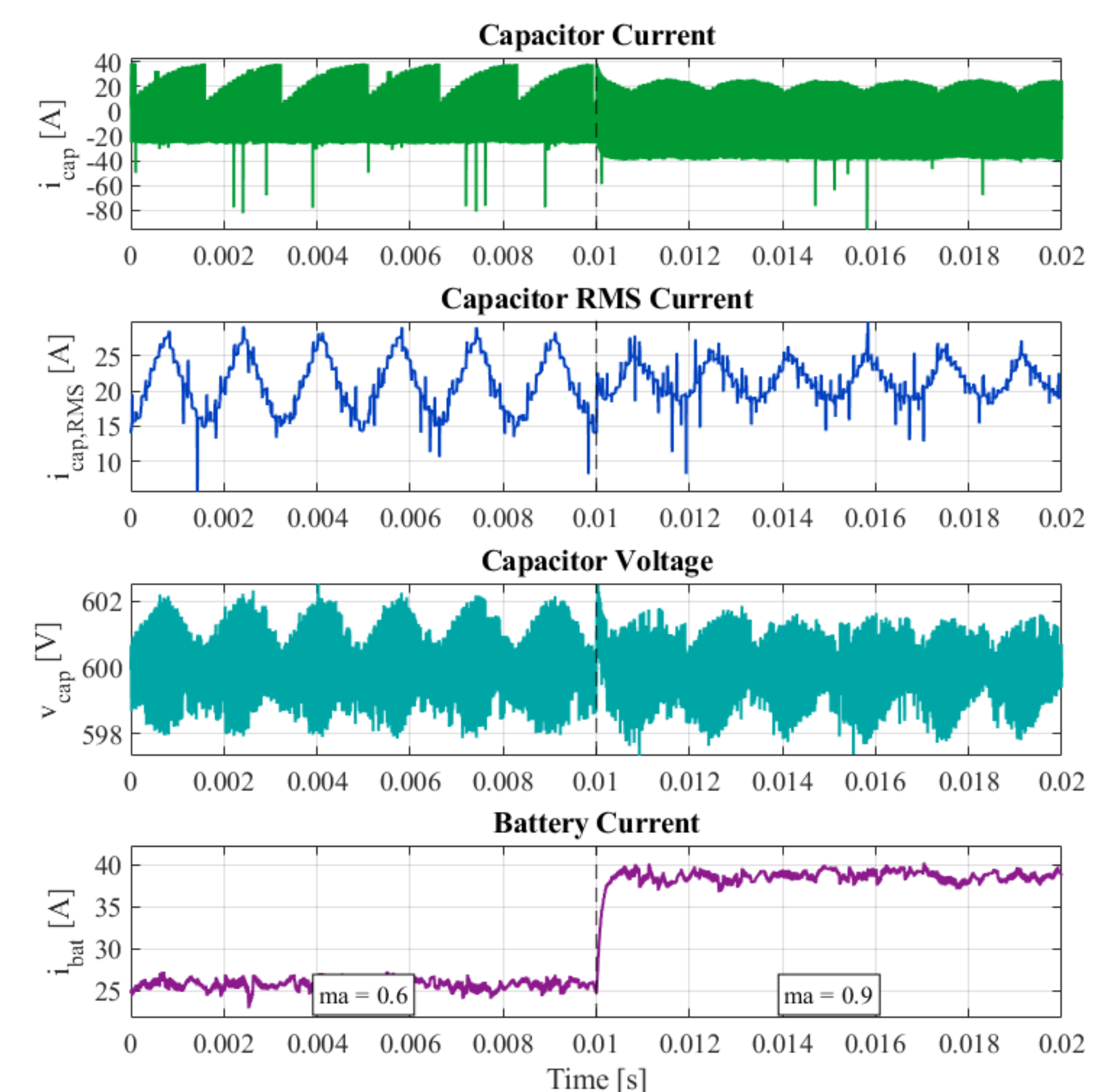
Hotspot temperature rise:

$$15.2^\circ\text{C} \quad 9^\circ\text{C}$$

Reduction ~40%

Estimated lifetime extension: ~50 %

Step change from $m_a = 0.6$ to $m_a = 0.9$



Transient performance:

- Smooth adaptation of carrier phase shifts.
- Confirms real-time feasibility.

9 CONCLUSION

- A novel adaptive multi-carrier PWM method is proposed for a single two-level three-phase inverter.
- The method requires no additional hardware and uses a reduced-dimension LUT for real-time implementation.
- The results show consistent DC-link capacitor RMS current reduction, improved thermal performance, and extended capacitor lifetime.

23% **case study** 27.6 A → 21.2 A
~50% **lifetime gain** example estimate

up to
27%
lower RMS current